Project 01

FQ Lab

Interaction product design / Gamified Education

BRIEF

Inspired by kids' tendency to **overspend** and the **gaps in financial education**, this project is designed to **develop children'skill** in daily money awareness, budgeting, allocation, and emergency planning through **scenario-based experiences**, making financial literacy fun and practical.

INDIVIDUAL PROJECT

DURATION

Dec. 2024 – Jun. 2025

BACKGROUND



FQ (Financial Quotient) is an important ability to understand and manage money for kids.

Nowadays, parents **ignore** financial intelligence education, and schools **do not have relevant courses.**

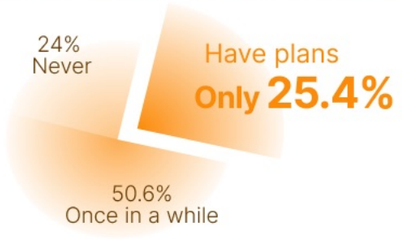
As a result, children's financial intelligence education is **far from the level it should be.**

RESEARCH

Financial Education Situation of Kids、Parents and Society

Kids

Do kids plan for spending money ?



The development of kids' thinking



	Age
Complex financial concepts	12
Be able to save money by budgeting	9
Understand delayed gratification	6
Identifying coins and denomination	3

Without financial education



About **50% of** those who did not receive financial education facing debt problems as adults.

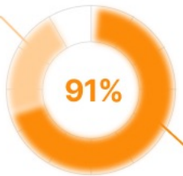
Children's financial literacy **lags behind** the financial realities they are exposed to.

Parents

Know the importance but have no idea.

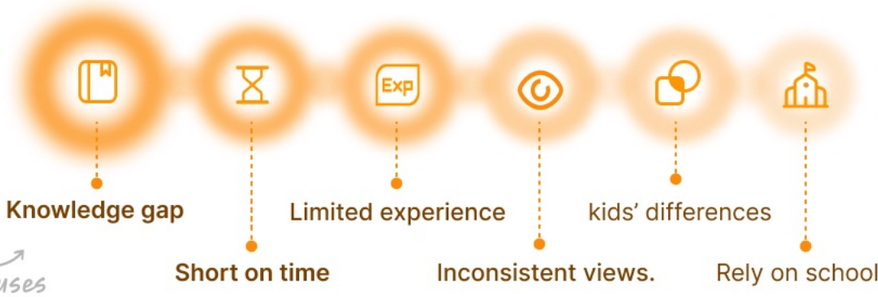
91% of parents recognize the importance of children's financial education.

Only 21% of parents have effective teaching experience.



Most parents care a lot, but don't know how to do.

Causes



Parents recognize the **importance of financial education** but **face many challenges** in implementing it.

Teachers & Society

Most schools lack financial literacy curriculum.

K12 Financial education is almost **non-existent in K-12**, with only a brief mention in high school math.



Children's financial literacy is influenced by **various society factors.**

Regional development, economic, the spread of social media... ...

Society **lacks** proper financial education for children, and **a systematic approach is yet to be established**.

Current Method of Financial Education

Curriculum
- Pros: Systematic and professional
- Cons: Theory-heavy and lack practicality

Picture book
- Pros: Fun and easy to understand
- Cons: Too simple, needs guidance

APPs
- Pros: Simple and easy to understand
- Cons: lacks flexibility for advanced management



brain-teasing toys
- Pros: promote creativity, and hands-on abilities.
- Cons: lack long-term appeal

piggy bank
- Pros: Simple and easy to understand
- Cons: Too simple, needs adult guidance

Real-life practicing
- Pros: Apply theoretical knowledge to real-life
- Cons: Limited by resources and environment

DESIGN GOALS

Build a diversified family financial **education system** that **enhances children's financial literacy**

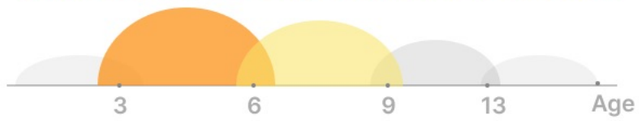
- Provide easy-to-implement methods
- Bridge the gap between theory and practice
- Move beyond one-size-fits-all education

QUESTIONNAIRE

Self-assessments by respondents

- Children's **interest** in financial literacy. 3.59
- Parents need **more knowledge.** 3.71
- Current education has **gaps**? 3.28
- Child's current financial literacy **level** 2.16

The ideal age to carry out financial education.



The methods parents typically used.



Parents' preferred educational approaches.

Most parents prefer to cultivate their children's financial awareness through **family-based education.**

Varying levels of parental involvement.



38% favor **joint parent-child** participation



47% prefer a **child-led approach** with **parental guidance.**

Curiosity of children. Parents' expectations.

Curiosity of children.	Parents' expectations.
Smart shopping	Learn Rational Consumption
Ways to make money	Develop the habit of saving
The price variation	Value and sources of money
Electronic Payment	Know marketing strategies

How Children Respond to and Engage with Different Forms.

I observed parent–child interactions during shopping and the process of children learning financial literacy.



No feedback ⇨ fading interest ⇨ unrealistic views

Location-dependent.

Overly simplistic knowledge transmission, failing to bridge theory & practice.



Requires parents' full-time accompaniment and involvement.



Risk of eye damage and digital dependency.

Does nothing more than store coins - can't teach real skills.



INTERVIEW



The Wang Family

- **Experience** in financial education:
- Child's financial education **level**:
- Current **methods** of education:

" We want to enhance children's financial awareness through **practical operation and Interesting ways of interaction** rather than theoretical teaching. "

" I hope I can **earn some money** and learn **how to grow it and use wisely**."



Laura (an 7-year-old kid)

- **Experience** in financial education:
- Laura's financial **level**:
- Current methods :

" I just want money to **come from my hard work**, and I'd llike to know how to **get stuff when it's on sale,** I want to save money."

" I don't want my parents just talking at me. Can't we **play some fun games to learn** how to handle money instead?"



Prof. Huang

- **Experience** in financial education:
- Main **approaches** used in education:
- **Founder** of an institution for early education.

" I prefer that parents should focus **less on one-way knowledge delivery, and maybe more on interactive experiences** with their children. "

" To understand **personalities** of individuals and gradually **nurture financial literacy**."



Ms. Li

- **Experience** financial education:
- Child's financial **level**:
- Current methods :

" I'm concerned about my child's **impulsive spending**, and want to help them understand the **relationship between cash and digital payments.**"

" I prefer guiding them through **simple daily behavior**. gradually tell them **money management and financial concepts.**"

PERSONA

Age **35**
Senior Management
Income：Middle to High

Feature

- Busy with work
- Values Education
- Progress together



Age **6**
Primary school
Level: basic

Feature

- Be curious
- Interest in financial
- Has pocket money

" Seeks **simple, easy-to-use and flexible** financial education approach that helps children build saving, spending habits **through practice.** "

" I Hope to learning money management through **interactive, play-based experiences** with proper **parental support.** "

Pain Points and Needs

	Engagement	Guidance	Time	Practicality
P	Financial learning feels dull and unmotivating.	Needs easy-to-follow guidance tools.	Limited time for deep parent-child interaction.	Needs tangible financial practice.
N	Needs fun, interactive ways to stay interested.	Parents unsure how to teach or guide properly.	Needs quick, efficient learning activities.	Concepts feel vague and hard to apply daily.

ANALYSIS

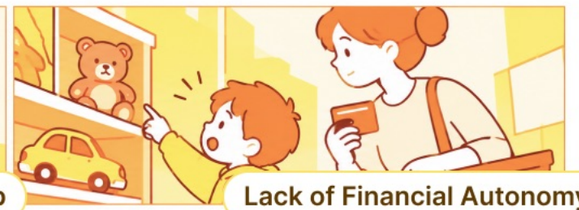
Describe & Empathize | Requirement | Opportunity | Innovation

Phenomenons

Early Financial Exposure



Knowledge–Practice Gap



Lack of Financial Autonomy



Lack of Financial Education



Misunderstanding of Money's Value

Requirement

- Easy-to-use home tools
- Task-driven interaction
- Age-appropriate education.
- Real-life experience training.
- Delayed gratification.
- Multidimensional expansion
- Money awareness: planning, choices, consequences.
- Low intervention, strong feedback and companionship.

OPPORTUNITY

Family financial micro-tasks
3-5 minutes each, tied to daily life, with clear feedback and rewards.



Parent-child collaborative interaction
completing tasks or discussing financial decisions together to better understand and practice financial literacy.



Integrate real-life scenarios into educational tools
Use real-life scenarios for kids to simulate shopping, investing... ...



Hybrid digital-physical solution
combining online and offline experiences for real and virtual feedback.

MINDMAP

I extend outward from the design opportunity points to obtain more specific methods that interest me to form the final output by making mindmap.



Definition

- Extract actionable solutions based on the mind map.

Financial skill acquisition 01
Impart practical skills rather than merely knowledge.

Scenario simulation 02
Avoid having knowledge that is too vague, dull and tasteless.

Human-computer interaction 03
Transform education into a intuitive and efficient experiential process.

- Solution methods

"AR Exploration" | "Behavior-Sensing Projection" | "Tabletop Game"

Using immersive projection to create life-like scenarios, where children can practice smart decision-making via intuitive behavioral interactions.

DESIGN PROCESS

Moodboard

Designed for children in family environments,
the product and interface feature **smooth shapes, gentle textures, and muted colors.**



Relevant Information

Based on the data of children, the basic size of the product, as well as its interaction range and content can be determined.

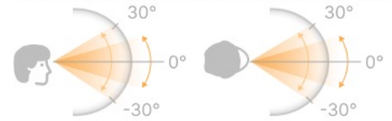
Hand measurement



Age	Circumstance	Length	Thumb
Baby	4.5	4.5	1.5
Toddler	5.5	5.5	1.75
Child	6.5	6.5	2
Woman	7.5	7.5	2.25
Man	8.5	8.5	2.5

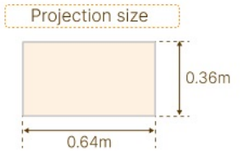
Range of visibility

To find the prime zone for key interactive area.



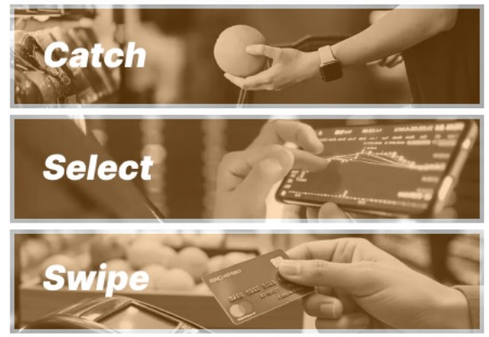
The central area of the visual field where information is perceived with the greatest clarity.



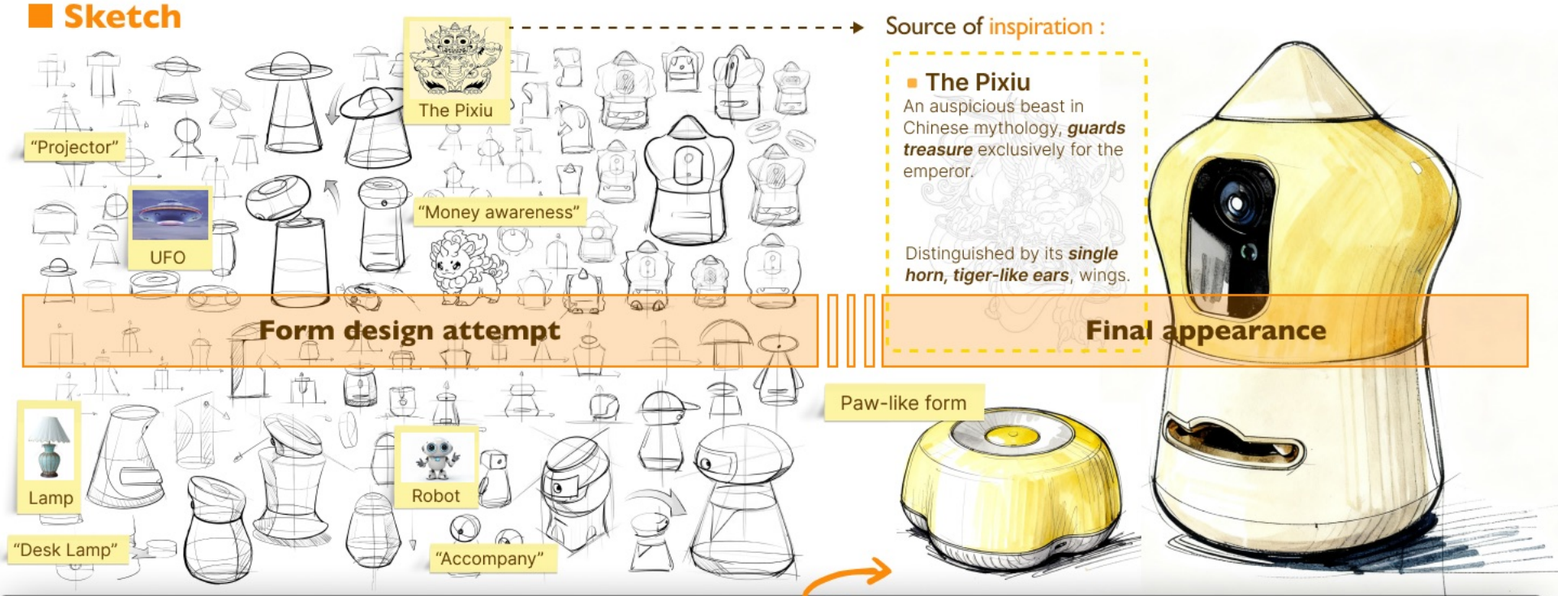


Actions

To find the usual action during financial activities



Sketch



Source of inspiration :

The Pixiu

An auspicious beast in Chinese mythology, ***guards treasure*** exclusively for the emperor.

Distinguished by its ***single horn, tiger-like ears***, wings.

Form design attempt

Final appearance

Human-Product Interaction Test

Test method

Through a **mini participatory workshop** with **low-fi clay prototypes**, Kids and I co-explored product form and usage with our target users.



Form exploration 01



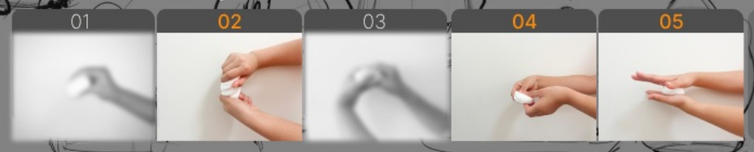
Form exploration 02

Exploring the most suitable shape for a comfortable grip.



Interection

Kids decide how they want to use.

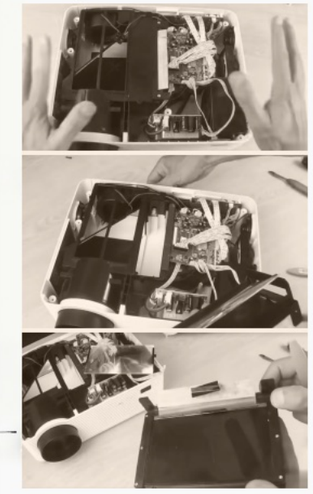


02,04 and 05 are selected by most of kids.

Modelling

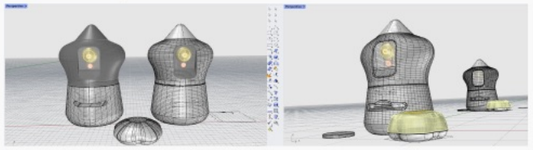
The product model was built based on a structural study, followed by 3D printing, prototyping, and physical assembly of the paw-shaped handle.

Projector Teardown

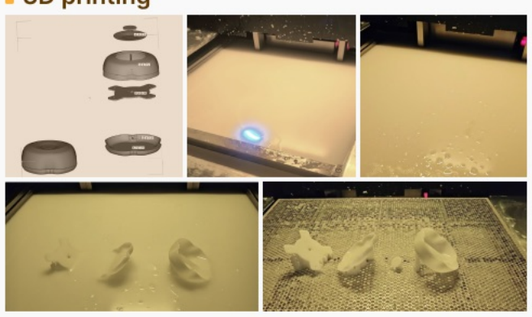


Using core layout for miniaturizing the product body.

Modeling with Rhino



3D printing



- Using SLA printing for casing and load-bearing parts
- Translucent resin replaces silicone for cost efficiency.

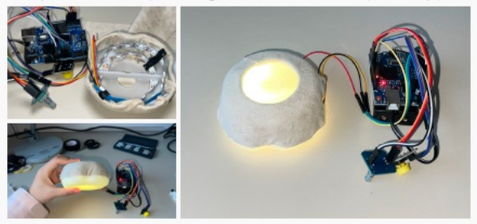
Assembly



- Making the suede casing and Arduino hardware.



- All parts were put together to form a prototype.



FINAL DESIGN

Product Display

FQ (Financial Quotient)

FQ Lab • Kids' contextual FQ skill-building system.



Types of coins

I break down daily expense into **five kid-friendly categories** for easy learning.



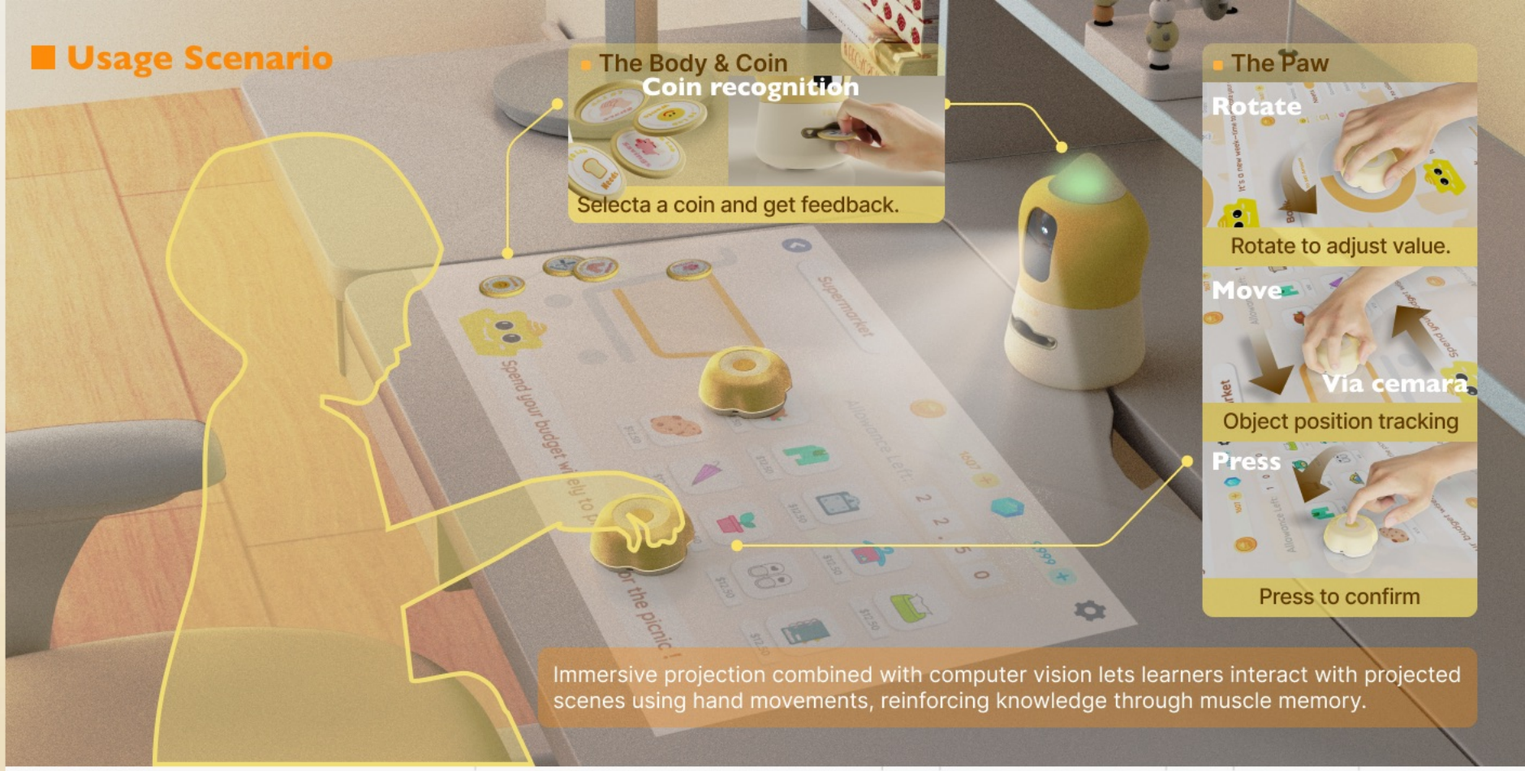
Element

FQ Lab comprises multiple modules:

- The main product combining input and output;
- Two palm-sized interactive units for parent-child use;
- Five types of pre-loaded category tokens.

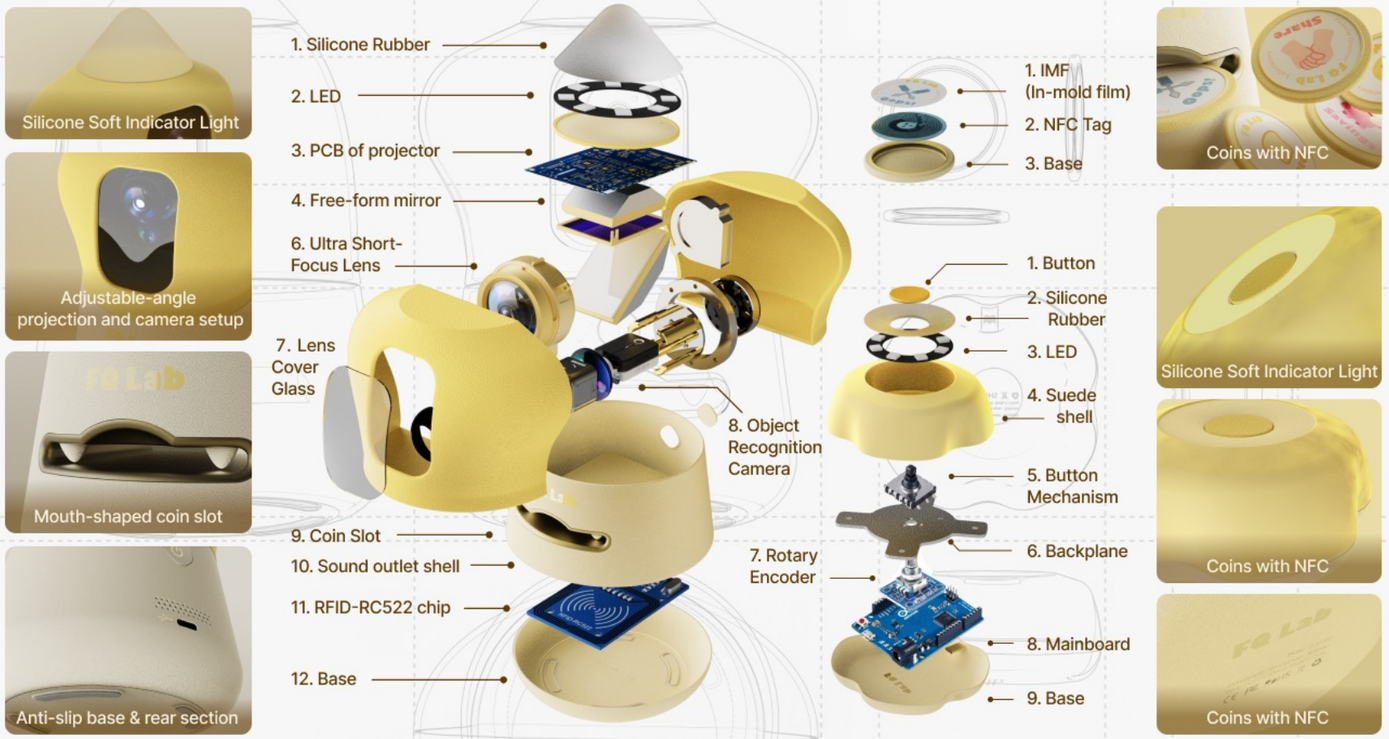


Usage Scenario



Structure and CMF

- The main unit integrates a short-throw projector, camera recognition system, LED feedback lights, and Close-range signal recognition;
- The interaction modules support pressing, rotating, and haptic feedback.
- Expense-category Coins are recognizable and classifiable.



- The ratio between information output and children's level of acceptance is relatively low.

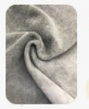
#F6E171 #F3E9C7



PLA
Biodegradable and non-toxic for kids.
With **durable grip** texture



Platinum Silicone
Certified-safe, Light-diffusing.
Matte-textured finish



Faux Suede
Soft and skin-friendly.
Sueding finish

UI & USAGE PROCESS

User Flow

Start → Selected scenes → Scenario learning & Feedback → Random Event → End

UI

Start Screen



1 Start the product and software system.

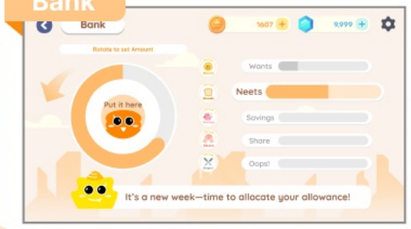
Map



Market · Bank · Park · Repair · School

2 Select the scenes they want to learn more.

Bank



3 Complete the financial tasks as instructed.

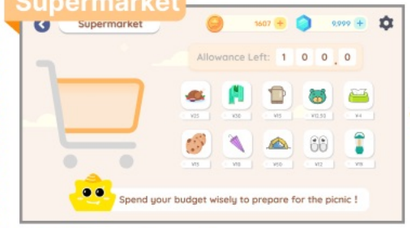
Adjust

Rotate to modify how money is distributed, practicing **budgeting skills**.

Repeat with feedback to **enhance financial skills** and **embed financial literacy naturally**.

4 Provide encouragement & guidance

Supermarket



Confirm · Select

Move and confirm to complete tasks within the scenario.

Feedback



Emergency



5 Emergency develop abilities to **cope with risks.**

Needs · Savings · Share · Wants · Oops!

Result Confirm





Select the **type of expense** by inserting coins into the body.

Final



Gain knowledge and hands-on money management skills to progress through levels.

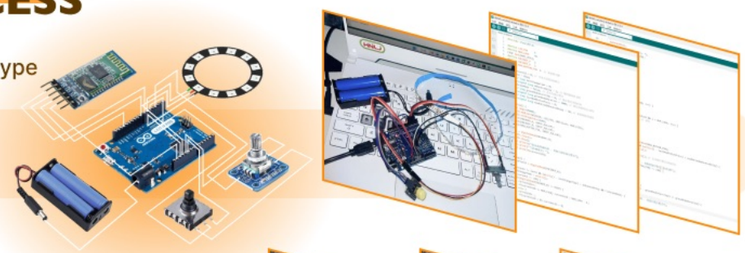


6 Acquire both knowledge and skills

REALIZATION PROCESS

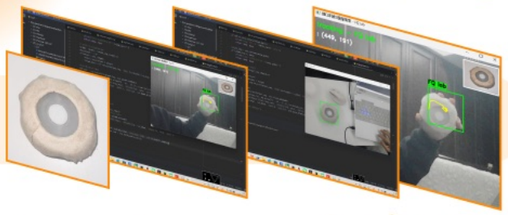
- The hardware interaction prototype was built by using ***Arduino.***

Using an Arduino board equipped with an **EC11 rotary encoder, buttons, an HC-05 Bluetooth module,** and **WS2812B LED**



- **Code validation**

Code worked well across various environments and backgrounds.



- ***CV-based*** Real-time Target Detection and Tracking by **Python.**

- Intergration with ***unity***

2D (Built-in Render Pipeline) to create UI effects





Using an external projector and camera to simulate the using process.

USER EXPERIENCE

- An 8-year-old child and her mother were invited to participate in test.



01 Contextual skills training

02 Decide and Get feedback



03 Parent-child co-learning

- **User feedback**



Rating: **(total 10)**

Fun	Learning	Interaction	Bonding
7	8	7	6

Want to say...



"I wish there were more things to do, like earn coins. Also, it was a bit sad that my hand would cover the images from the projector."

REFLECTION

In what aspects does this FQ skill-building system still need improvement?





FQ Lab still has many areas for improvement: It **cannot fully simulate real-life situations**, and **competitive elements** can be added in parent–child interaction, also, it cannot fit all families due to **different values and spending habits**.